Comparing Data-Driven and Physics-Informed Neural Networks

Main finding: the physics-informed model had slightly lower test error. Both models have limited original-scale accuracy; this comparison does not isolate the physics penalty's effect or validate the physical law.

1. Data and evaluation design

Item	What we did
Dataset	1,865 observations; both prediction networks use all nine features, x1–x9. ID is only an identifier.
Data quality	No missing/nonfinite values or duplicate IDs. All observations, including extreme values, were retained.
Repeated inputs	Six matching-input pairs: two share y, four have different y. All rows were kept; each pair stays together in every split.
Holdout split	Seed 42; 1,493 training / 372 testing rows (80.05% / 19.95%). Grouping explains the small departure from 80/20.
Validation	The same five grouped folds for both models; validation sizes 298, 298, 300, 300, 297. Preprocessing is fitted on each fitting fold only; the test set is excluded from tuning.

Preparation: compare raw inputs with log10(x1, x2, x4, x5, x6), then standardize. These inputs are positive; zero-valued features are not logged. The target is log10(y), reflecting its wide range (about 0.000075–46.86). Outputs are converted back with 10^z, ensuring positive predictions. Invalid numerical outputs raise errors; nothing is silently clipped.

2. Search choices and selected models

Search choice	Candidate values	Reason
Prediction network	1–3 layers; width 16/32/64; ReLU/tanh	Compare modest capacities for nine inputs.
Adam learning rate	0.0003 / 0.001 / 0.003	Explore slower and faster updates.
Batch size; L2	32/64/128; 0/1e−5/1e−4/1e−3	Balance update frequency and overfitting.
Input representation	Selected logs off/on	Test compression, especially for x2 and x4.
Exponent network	Tanh: (8), (16), (16,8); inputs x1–x2/all	Limit flexibility; test additional inputs.
Physics weight λ	0.01 / 0.1 / 1 / 10	Compare weak and strong equation penalties.

Budget: seed-42 random sampling without replacement; 20/1,296 data-driven and 20/72 physics candidates, each over five folds (200 fits). Physics keeps the selected prediction backbone, batch size, L2, and input transform fixed. Choose the lowest mean validation RMSLE. Early stopping: at most 1,000 epochs; patience 50; minimum improvement 1e−5. Retain the lowest-score checkpoint. L2 penalizes weights, not biases; no separate weight decay is used.

Selected setting	Data-driven	Physics-informed
Prediction network	16 → 16 → 16, ReLU	Same prediction backbone
Rate / batch / L2	0.003 / 64 / 0.001	0.003 / 64 / 0.001
Transforms	Selected logs + standardization; log10(y)	Same
Exponent / physics weight	Not applicable	One 16-neuron tanh layer; x1–x2; λ=0.01

3. How physics enters training

The equation is $y_{phys} = x2^2 / [72.2 \times x1^{(-m)}]$. A second network learns a separate m for each sample, jointly with the prediction network. Softplus enforces m>0. At inference, m uses features alone, never observed y. Features x3–x9 influence the prediction branch.

Using original units: $z_{phys} = 2\log10(x2) - \log10(72.2) + m\log10(x1)$, where z = log10(predicted y). Minimize data MSE in log10(y) + λ × mean$[(z - z_{phys})^2]$ + L2. Equation equivalence and gradient flow to both networks and m were checked. Exponents are learned during training, not prescribed or fitted afterward.

4. Training and validation evidence

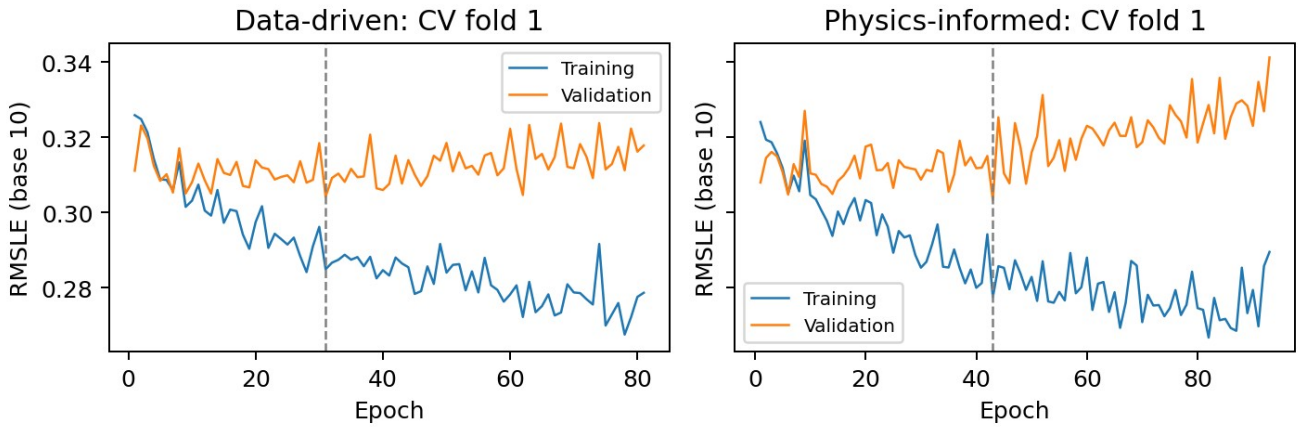


Figure 1. Training and validation RMSLE for preselected fold 1 of each selected model. Dashed lines mark retained epochs. Curves stop at actual early stopping. All five folds are included in supporting outputs; test error is never plotted against epoch.

Selected model	Validation RMSLE mean	Fold standard deviation
Data-driven	0.299685	0.016474
Physics-informed	0.298574	0.015089

Each summary uses five folds; SD uses ddof=1. These scores were used for selection and are not independent test estimates. Fold seeds are 42 + fold number.

5. What the exponent learned

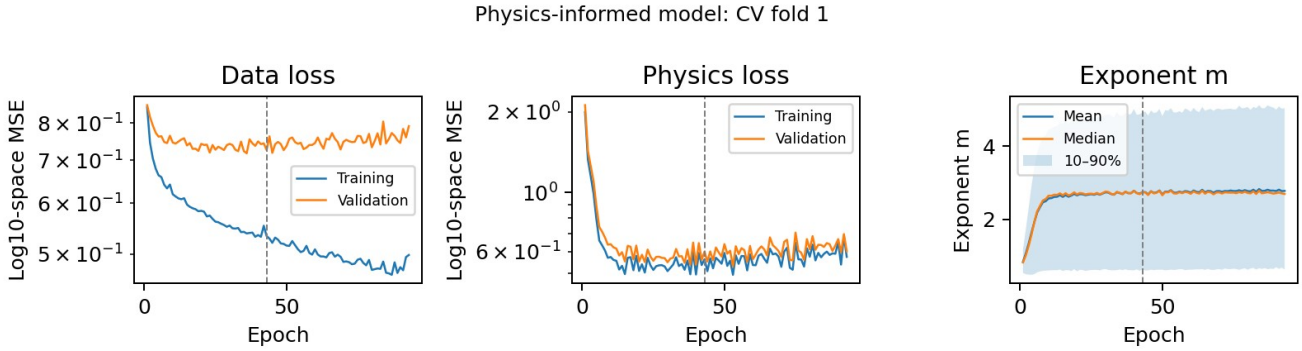


Figure 2. Physics-informed fold 1: separate data/physics losses and m evolution on fixed fitting-fold inputs. The shaded band is the sample 10th–90th percentile, not uncertainty. Dashed lines mark the retained epoch. Final-refit diagnostics are also saved.

Because m varies by sample, report its final distribution rather than one global value. Each sample's learned m is saved with its ID; held-out values are generated from inputs alone.

Final model inputs	Mean m	Median m	SD	10th–90th percentile
Training	2.843	2.718	1.723	0.649–5.296
Testing	2.959	2.958	1.652	0.768–5.419

6. Runtime and final refitting

Model	100 CV fits	Full-training refit
Data-driven	293.20 s	78 epochs; 1.29 s
Physics-informed	546.87 s	116 epochs; 2.67 s

About 14 minutes of CV training on one CPU thread, using deterministic algorithms; installation and smoke tests excluded. One nonselected physics-informed fold reached 1,000 epochs; selected folds did not. Refit durations are the median selected CV best epochs. Fresh models and scalers use all 1,493 training rows, seed 42, with no validation/test monitoring. Both models were fixed before testing and were not adjusted afterward.

7. Final predictive results

Model	Training R^2	Training RMSLE	Testing R^2	Testing RMSLE
Data-driven	0.0896	0.2808	0.1803	0.2855
Physics-informed	0.1886	0.2725	0.2082	0.2810

Higher R^2 and lower RMSLE are better. $R^2 = 1 - SSE/SST$ uses original-scale residuals, not squared correlation. Base-10 RMSLE = $\sqrt{\text{mean}[(\log10(1+\hat{y}) - \log10(1+y))^2]}$. This +1 convention differs from the log-target training loss.

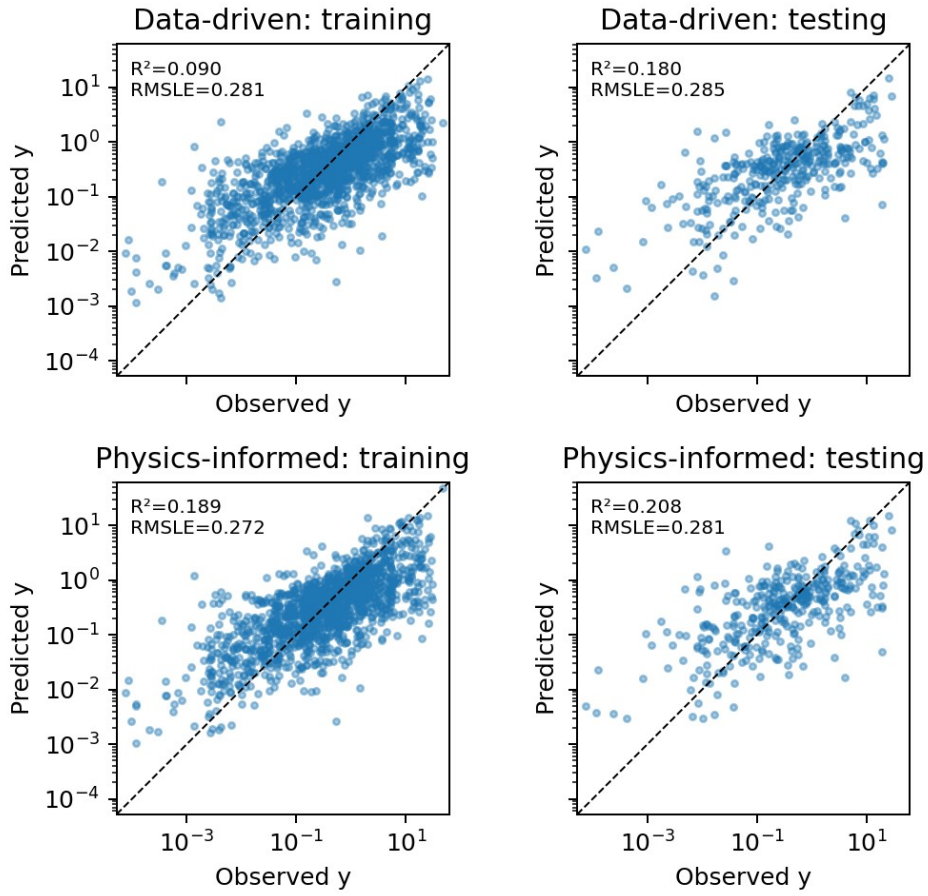


Figure 3. Final predictions: training and testing for both models. All panels use common logarithmic limits and a 1:1 line. Training predictions are in-sample; testing predictions are held out.

8. Interpretation and limitations

- Prediction limits: the physics-informed model has test RMSLE 0.2810 versus 0.2855, but both test R^2 values remain low. Small targets are often overpredicted and large targets underpredicted. The log(y) training loss and log(1+y) selection score emphasize errors differently; one split gives limited evidence.
- Approximate physics: agreement RMSE is 0.7838 (training) and 0.7075 (testing), in log10 units. The selected $\lambda=0.01$ is weak. Since $x1<1$ and $m>0$, the equation requires $y_{phys}<x2^2/72.2$; 359/1,493 training observations exceed that bound and were retained.
- Evidence limits: m and the predictor can co-adapt; the added branch and optimization changes are not isolated, so a causal benefit of physics is not established. Twenty candidates and one seed per fold limit robustness; fold SD is not a confidence interval. Matched-λ studies, repeated seeds, and simple baselines remain future work.

Reproduction and disclosure

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